

VANSHAJ BINDAL

Google Summer of Code 2025 @ QuTiP | MSc Physics Graduate

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EDUCATION

MSc in Physics

Cardiff University

-  09/2023 - 09/2024  Cardiff, UK
- Student representative for the program
 - Award Classification: 1st/ Distinction
 - Concentration: Theoretical Physics, Condensed Matter Theory, Quantum Information
 - *Relevant Coursework:* Quantum Theory of Solids, Advanced Particle Physics, Data Analysis, Statistical Mechanics, Theoretical Physics, Philosophy of Physics
 - Thesis: Holographic Quantum Error Correction, Supervisor: Dr Pieter Naaijken, Score: 83%

PG Certificate in Physics

Queen Mary University of London

-  09/2021 - 09/2022  London, UK
- Award Classification: 2.1/Merit
 - Concentration: Theoretical Physics, High Energy Physics
 - *Relevant Coursework:* Quantum Mechanics, Spacetime & Gravity, Solar System, Intro QFT, Adv QFT, Intro String Theory

BTech in Industrial and Production Engineering

Manipal Institute of Technology (MIT Manipal)

-  08/2016 - 02/2021  Manipal, India
- Concentration: Mechanical Engineering, Applied Physics, Material Science
 - *Relevant Coursework:* Math-4, Fluid mechanics, Radiation Physics, Engineering Physics, Physics lab, Basic electrical technology etc
 - Thesis: Experimental & Data analysis on Fiber Metal Laminates - Scored a perfect 10 GPA

EXPERIENCE

School on Quantum Dynamics of Matter, Light and Information

Abdus Salam International Centre for Theoretical Physics

-  08/2025 - 09/2025  Trieste, Italy
- Exact Dates: 18 Aug - 05 Sep, 2025
 - Awarded competitive ICTP grant covering full accommodation and meals for 3-week intensive program, selected from international applicant pool to study advanced quantum many-body physics and computational methods
 - Completed 14 specialised courses (42 sessions total) covering cutting-edge topics including tensor networks, topological phases, quantum circuits, and Markov-chain methods taught by world-leading researchers from Harvard, MIT, Oxford, and other top institutions
 - Gained hands-on expertise in advanced computational techniques for quantum systems, particularly in tensor network algorithms and scientific Python programming, directly applicable to quantum simulation and many-body physics research

Google Summer of Code 2025

QuTiP (NumFOCUS)

-  05/2025 - Present  Remote
- Selected as a Google Summer of Code 2025 contributor to work on the Hamiltonian Library for QuTiP
 - Currently developing a new quantum systems module which would include models such as Jaynes-Cummings, Rabi, Dicke, and various spin chain models for ready to use implementation
 - Designing a standardised API to support JSON serialisation, time-dependent parameters, and LaTeX representation
 - Collaborating with the QuTiP developer community to iteratively test and refine features, contributing to the open source ecosystem
 - Please visit my [gsoc-blog](#) for regular updates

SUMMARY

MSc Physics graduate (Distinction, Cardiff University) with specialised expertise in quantum information theory and theoretical physics. Currently contributing to QuTiP as a Google Summer of Code developer, building a quantum systems library. Proven track record securing competitive funding (ICTP grant, Cardiff scholarship) and strong technical skills in Python, tensor networks, and quantum simulation frameworks. Active contributor to quantum computing research community through software development, academic research, and educational outreach.

SKILLS

Python (Advanced)		LaTeX	Linux
GitHub	Julia (Beginner)		QuTiP
Qiskit	NumPy	Pandas	SciPy
Matplotlib		Machine Learning (Beginner)	
Tensor Networks		Monte Carlo Methods	
Quantum Computing		Theoretical Physics	

INTERESTS

-  Quantum Technologies & Computing
-  Open Source Scientific Software
-  Scientific Education & Outreach
-  Philosophy of Physics
-  Quantum foundations

KEY ACHIEVEMENTS

-  **ICTP Grant**
Selected for prestigious ICTP grant covering full accommodation and participation in elite 3-week physics program
-  **Cardiff University Scholarship for master's programme - 2023**
Awarded £5000 towards my masters program
-  **Student Leadership**
Selected as student representative for MSc Physics program at Cardiff University
-  **Distinction In MSc**
Achieved a Distinction classification with a score of 83% for MSc thesis.

EXPERIENCE

Mentee/Research Student

QOSF - Mentorship Program - Cohort 10

📅 10/2024 - Present 📍 Remote

- Selected for the quantum computing mentorship program run by Quantum Open Source Foundation (QOSF)
- As part of the screening task I developed a simulation framework for quantum circuits using matrix and tensor-based approaches, analysing runtime and scalability - [repository](#)
- Currently undertaking a project on classical simulation of quantum systems. This work involves implementing real and imaginary time evolution algorithms in the Pauli basis and developing efficient methods for Heisenberg picture simulations

Volunteer

XPANSE - Exponential Technologies Summit

📅 11/2024 - 11/2024 📍 Abu Dhabi, United Arab Emirates

- Selected as a volunteer for this global summit, focused on advancing cutting-edge physics, technology and multidisciplinary innovation.
- Assisted in organizing and managing sessions featuring prominent physicists, scientists including keynote speeches, panel discussions, and networking events.
- Facilitated smooth communication and coordination between speakers, attendees, and the event management team.
- Gained firsthand experience in event logistics and fostered professional connections within the physics and technology communities.

Participant

IQOQI - Summer School

📅 09/2024 - 09/2024 📍 Vienna, Austria

- Attended a rigorous week-long program on quantum optics and information science, hosted by IQOQI in Vienna and Innsbruck, Austria
- Participated in lectures by eminent researchers on topics spanning quantum mechanics, computing, and information theory, deepening expertise in both theoretical and experimental aspects of quantum science
- Explored cutting-edge experimental research facilities through lab tours, gaining hands-on insight into the practical implementation of quantum technologies

Research Student

Condensed Matter Group - Cardiff University

📅 10/2023 - 03/2024 📍 Cardiff, UK

- As a research student, conducted a focused theoretical and computational study on Quantum Electrodynamics (QED) to investigate dissipative processes within the Jaynes-Cummings model, under the guidance of Dr Amy Morreau
- Developed and ran simulations using the QuTiP software to model decoherence and quantum noise effects
- Built custom functions for better utility while running simulations
- Authored a comprehensive report detailing the methodology, findings, and implications of the study, showcasing ability to synthesise complex information into a professional report - [repository](#)

Teaching Assistant

Cardiff University

📅 09/2023 - 12/2023 📍 Cardiff, UK

- I was teaching assistant for the module "Physics of Fields and Flows"
- Primary responsibility was marking papers and homework and tutoring students

Research Student

Monte Carlo Methods in Statistical Field Theories

📅 07/2023 - 08/2023 📍 Remote

- Selected for a rigorous two-month project under the mentorship of UC Berkeley faculty
- Focused on study of statistical mechanics and statistical field theories
- Learnt to apply elementary Monte Carlo methods in context of SFT's
- Prepared and presented a final report and a poster

REFERENCES

Dr. Pieter Naaijken

NaaijkenP@cardiff.ac.uk

Dr. Amy Morreau

MorreauAl@cardiff.ac.uk